

SECURITY ASSESSMENT REPORT – 06

SSH Authentication Monitoring and Log Analysis using Splunk

Kali Linux + Metasploitable 2 + Splunk Enterprise

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1. Introduction

This assessment focused on security monitoring and authentication-log analysis rather than exploitation. A controlled SSH authentication scenario was created between Kali Linux and a Metasploitable 2 target. Failed and successful SSH authentication events were generated and examined in the target's authentication log. The resulting auth.log file was then imported into Splunk Enterprise and searched using Splunk Processing Language (SPL). The objective was to understand how raw operating-system security events can be collected into a centralized log-analysis platform, filtered for relevant activity, and reviewed as a sequence of authentication events. The Ubuntu documentation identifies /var/log/auth.log as a log location containing authorization-related activity such as password prompts and remote logins.

2. Objectives

The assessment was designed to identify the SSH service on the target, generate controlled failed and successful authentication activity, inspect the resulting authentication records, ingest the records into Splunk, and perform focused searches for failed and successful authentication events.

3. Laboratory Environment

Kali Linux: security testing and analysis host, observed address 192.168.202.131.
Metasploitable 2: controlled vulnerable target, observed address 192.168.202.130.
SSH: remote authentication service exposed on TCP port 22.
Splunk Enterprise: local log-ingestion and analysis platform.
Authentication log: /var/log/auth.log.

4. Methodology

The first stage established that the target was reachable and that SSH was exposed on TCP port 22. The SSH service was then used to create controlled authentication activity. Several unsuccessful authentication attempts were generated, followed by a legitimate successful login. On the target, the authentication log was queried to verify that both event types had been recorded. The authentication log was then made available to Splunk Enterprise. Splunk Web was used to upload the log, after which the data was searched using the source field and the linux_secure sourcetype. Separate searches were performed for failed and accepted passwords. A combined search was finally used to display both event types in one result set.

5. Evidence and Results

```
(kali㉿kali)-[~]
$ sudo nmap 192.168.202.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-30 06:41 -0400
Nmap scan report for 192.168.202.130
Host is up (0.0011s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:FA:DD:2A (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.72 seconds

(kali㉿kali)-[~]
$ sudo nmap -p 22 192.168.202.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-30 06:42 -0400
Nmap scan report for 192.168.202.130
Host is up (0.00072s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 00:0C:29:FA:DD:2A (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.66 seconds

(kali㉿kali)-[~]
$
```

Figure 1: Nmap reconnaissance showing TCP/22 open and SSH exposed on the Metasploitable target.

This established the network-level context for the subsequent SSH authentication activity.

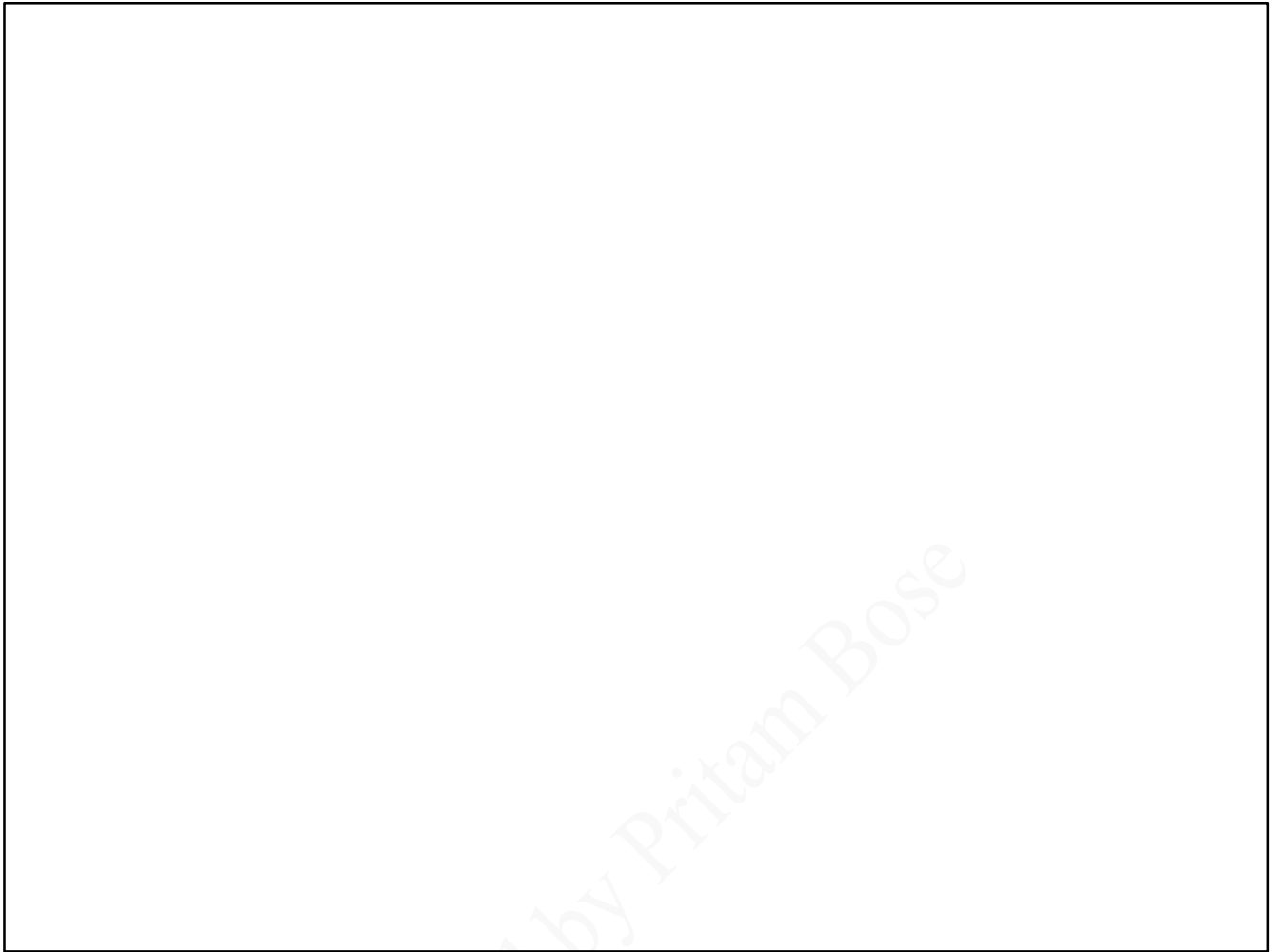


Figure 2: Metasploitable auth.log showing failed and accepted SSH authentication records.

The raw log contained the account, source address, timestamp and SSH protocol information, confirming the events existed before centralized analysis.

source="auth.log" "Failed password"

✓ 2 events (before 8/30/26 10:32:33.000 AM) No Event Sampling ▼

Events (2) Patterns Statistics Visualization

Timeline format ▼ - Zoom Out + Zoom to Selection X Deselect

Format ▼ Show: 20 Per Page ▼ View: List ▼

	Time	Event
>	8/30/26 9:36:43.000 AM	Aug 30 09:36:43 metasploitable sshd[8445]: Failed password for msfadmin from 192.168.202.131 port 42546 ssh2 host = kali source = auth.log sourcetype = linux_secure
>	8/30/26 6:51:53.000 AM	Aug 30 06:51:53 metasploitable sshd[7919]: Failed password for msfadmin from 192.168.202.131 port 33440 ssh2 host = kali source = auth.log sourcetype = linux_secure

SELECTED FIELDS

- a host 1
- a source 1
- a sourcetype 1

INTERESTING FIELDS

- # date_hour 2
- # date_mday 1
- # date_minute 2
- a date_month 1
- # date_second 2
- a date_wday 1
- # date_year 1
- a date_zone 1

Figure 3: Splunk search for failed authentication: source="auth.log" "Failed password".

The search isolated failed authentication events and demonstrated that the imported log was indexed and searchable.

New Search SPL Convert to SPL2

source="auth.log" "Accepted password"

✓ 2 events (before 8/30/26 10:35:38.000 AM) No Event Sampling

Events (2) Patterns Statistics Visualization

Timeline format Zoom Out + Zoom to Selection X Deselect

Format Show: 20 Per Page View: List

Hide Fields All Fields

SELECTED FIELDS

- host 1
- source 1
- sourcetype 1

INTERESTING FIELDS

- # date_hour 2
- # date_mday 1
- # date_minute 2
- # date_month 1

i	Time	Event
>	8/30/26 9:36:52.000 AM	Aug 30 09:36:52 metasploitable sshd[8445]: Accepted password for msfadmin from 192.168.202.131 port 42546 ssh2 host = kali : source = auth.log : sourcetype = linux_secure
>	8/30/26 7:01:19.000 AM	Aug 30 07:01:19 metasploitable sshd[8099]: Accepted password for msfadmin from 192.168.202.131 port 59966 ssh2 host = kali : source = auth.log : sourcetype = linux_secure

Figure 4: Splunk search for accepted authentication: source="auth.log" "Accepted password".

The search independently identified successful SSH authentication events in the indexed data.

New Search SPL Convert to SPL2

source="auth.log" "Failed password" OR "Accepted password"

✓ 4 events (before 8/30/26 11:05:18.000 AM) No Event Sampling

Events (4) Patterns Statistics Visualization

Timeline format Zoom Out + Zoom to Selection X Deselect

Format Show: 20 Per Page View: List

	i	Time	Event
SELECTED FIELDS a host 1 a source 1 a sourcetype 1 INTERESTING FIELDS # date_hour 3 # date_mday 1 # date_minute 3 a date_month 1 # date_second 4 a date_wday 1 # date_year 1	>	8/30/26 9:36:52.000 AM	Aug 30 09:36:52 metasploitable sshd[8445]: Accepted password for msfadmin from 192.168.202.131 port 42546 ssh2 host = kali ; source = auth.log ; sourcetype = linux_secure
	>	8/30/26 9:36:43.000 AM	Aug 30 09:36:43 metasploitable sshd[8445]: Failed password for msfadmin from 192.168.202.131 port 42546 ssh2 host = kali ; source = auth.log ; sourcetype = linux_secure
	>	8/30/26 7:01:19.000 AM	Aug 30 07:01:19 metasploitable sshd[8099]: Accepted password for msfadmin from 192.168.202.131 port 59966 ssh2 host = kali ; source = auth.log ; sourcetype = linux_secure
	>	8/30/26 6:51:53.000 AM	Aug 30 06:51:53 metasploitable sshd[7919]: Failed password for msfadmin from 192.168.202.131 port 33440 ssh2 host = kali ; source = auth.log ; sourcetype = linux_secure

Figure 5: Combined Splunk search showing failed and accepted authentication events together.

The combined result returned four relevant events. It showed failed and accepted password events associated with 192.168.202.131 and the msfadmin account. A visible sequence showed a failed authentication at 09:36:43 followed by an accepted authentication at 09:36:52.

6. SPL Queries Used

The following searches were used during the assessment:

```
source="auth.log" "Failed password"
```

```
source="auth.log" "Accepted password"
```

```
source="auth.log" "Failed password" OR "Accepted password"
```

The first two searches independently filtered failed and accepted password events. The final search combined the two conditions so both authentication outcomes could be reviewed together.

7. Analysis and Security Interpretation

The assessment demonstrates a basic security-monitoring workflow. An SSH service exposes a remote authentication surface, and authentication attempts generate records that can be examined after the fact. A single failed login does not prove malicious activity because it can result from an incorrect password. However, repeated failures followed by a successful login from the same source are security-relevant and deserve investigation in a real environment. In this controlled laboratory, the events were intentionally generated by the tester. They should therefore not be described as evidence of a real compromise or brute-force attack. The accurate finding is that the monitoring workflow successfully detected and correlated failed and successful SSH authentication events. The exercise also demonstrates the value of centralized log analysis. Instead of manually opening a log file and searching for strings, Splunk provides a searchable interface where events can be filtered and reviewed together. SPL is designed for retrieving and analyzing indexed events, making this workflow applicable to larger datasets.

8. Security Recommendations

SSH access should be restricted to accounts and networks that actually require it. Authentication events should be retained and monitored so repeated failures, unusual source addresses, and successful logins following suspicious activity can be investigated. Where practical, stronger authentication such as public-key authentication and multi-factor authentication should be considered. Accounts with unnecessary remote access should be removed or restricted, and privileged accounts should receive additional monitoring. For a production SOC workflow, this search could be extended into statistical queries, dashboards and alerts, then correlated with process creation, endpoint and network telemetry.

9. Limitations

This was a controlled laboratory exercise using Metasploitable 2 and deliberately generated authentication activity. The evidence demonstrates the detection workflow but does not represent a real-world intrusion. The dataset was small, so no statistical baseline or automated alerting threshold was established. The assessment focused on authentication-event visibility in Splunk rather than full incident response or endpoint compromise analysis.

10. Conclusion

The assessment successfully demonstrated an end-to-end security monitoring workflow for SSH authentication activity. The target's SSH service was identified through network reconnaissance, controlled failed and successful login attempts were generated, and the resulting authentication records were verified directly in `/var/log/auth.log`. The log was then imported into Splunk Enterprise, where targeted SPL searches were used to isolate failed authentication events, successful authentication events, and finally both categories together. The most important outcome was understanding the relationship between an operating-system event, the raw evidence generated by that event, and the centralized analysis performed by a security monitoring platform. The final combined search showed four relevant authentication events and made it possible to observe failed and accepted SSH activity from the same source address. In an operational SOC, this evidence could form the starting point for a larger investigation into whether the activity was expected, suspicious, or part of a broader attack sequence. The exercise provided practical exposure to log collection, authentication-event analysis, basic SPL searching, and security-event interpretation. Future work can extend the workflow with additional hosts and log sources, statistical detections, alerts, dashboards, and correlation with network and endpoint telemetry.

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11. References

1. **Splunk Documentation:** Search Reference – SPL syntax and commands. <https://docs.splunk.com/Documentation/SplunkCloud/latest/SearchReference/>
2. **Splunk Documentation:** Adding data through Splunk Web, including Upload. <https://docs.splunk.com/Documentation/SplunkCloud/latest/Data/Howdoyouwanttoadddata>
3. **Splunk Documentation:** Search command usage and search-term behavior. <https://docs.splunk.com/Documentation/SCS/latest/SearchReference/SearchCommandUsage>
4. **Ubuntu:** Viewing and monitoring log files; /var/log/auth.log. <https://ubuntu.com/tutorials/viewing-and-monitoring-log-files>
5. **Ubuntu Server Documentation:** OpenSSH server configuration and authentication methods. <https://ubuntu.com/server/docs/how-to/security/openssh-server/>
6. **Ubuntu Community Help:** OpenSSH logging and authentication-log guidance. <https://help.ubuntu.com/community/SSH/OpenSSH/Configuring>
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9. **MITRE ATT&CK:** DET0596 – Behavioral Detection of Remote SSH Logins Followed by Post-Login Execution. <https://attack.mitre.org/detectionstrategies/DET0596/>
10. **MITRE ATT&CK:** M1018 – User Account Management. <https://attack.mitre.org/mitigations/M1018/>
11. **NIST:** SP 800-92 – Guide to Computer Security Log Management. <https://csrc.nist.gov/pubs/sp/800/92/final>
12. **NIST:** Cybersecurity Log Management Planning Guide project. <https://csrc.nist.gov/Projects/log-management>

Note: This assessment was performed in an isolated laboratory environment using systems intentionally configured for security testing. No production systems were targeted.